EESSI

EUROPEAN ENVIRONMENT FOR SCIENTIFIC SOFTWARE INSTALLATIONS

3 July 2025

https://github.com/EESSI/meetings/wiki

Agenda

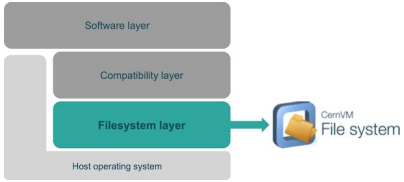
- Quick introduction by new people
- Progress update per EESSI layer
- Update on EESSI production repository `software.eessi.io`
- Update on EESSI test suite, build-and-deploy bot, documentation
- AWS/Azure sponsorship update
- EESSI governance
- Upcoming/recent events
- Q&A

Quick introduction by new people

New people on the call: feel free to introduce yourself!

- Who are you, where do you work, on what?
- Why are you interested in the EESSI project?
- Are you planning to actively contribute,
  and if so, to which aspect(s) of the project?

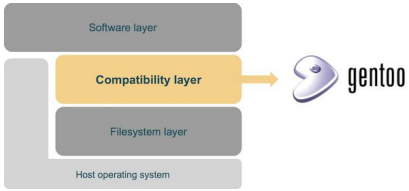
Progress update: filesystem layer



- All CernVM-FS servers have been updated to CernVM-FS 2.13.1
- **Stratum 0 now only accepts signed tarballs**
  - Tarballs that are not signed or cannot be verified will be skipped
- Nested CVMFS catalogs are being created for the accelerator (sub)stacks (PR #214)
- Build container images based on Debian 12 (PR #221)
  - Includes `unionfs` instead of `fuse-overlayfs` (PR #226),
    which doesn't suffer from the permission-related problems for software rebuilds

[Bob, Kenneth]

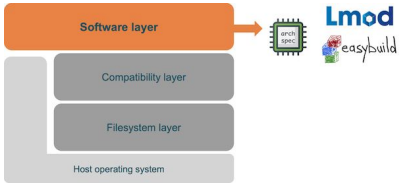
Progress update: compatibility layer



- Compatibility layer for 2025.06 has been built and ingested
  - Available for `aarch64`, **`riscv64`**, `x86_64`
  - `/cvmfs/software.eessi.io/versions/2025.06/compat/linux/`
  - PR #209 for playbooks & co for compat layer
  - PR #108 for package set
  - Due to an issue with user id resolving, it was rebuilt in PR #223
- Fully built and deployed by the bot
- Full discussion on new version in support issue #56

[Bob, Kenneth]

Progress update: software layer (1/2)



Split of software layer and software layer scripts! See next slides on this

Highlights of recently merged software PRs:

- MAQAO 2.21.1 (PR #1081), EasyBuild 5.1.0 (PR #1090), buildenv (PR #1096, PR #1101), OpenFOAM v12 (PR #1113), VSEARCH 2.30.0 (PR #1114), cutadapt 5.0 (PR #1116), MEGAHIT 1.2.9 (PR #1117), BWA 07.18 (PR #1123), Ginkgo 1.9.0 (PR #1127), HeFFTe 2.4.1 (PR #1130)

NVIDIA GPU software:

- PRs: CUDA+OSU-Micro-Benchmarks (#1030, #1076, #1077), CUDA-Samples 12.1 CC70 (#1083)

A64FX CPU target (via service account on Deucalion):

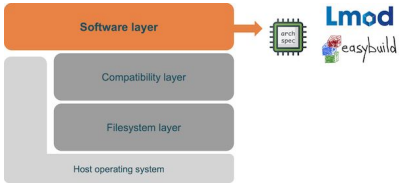
- PRs: #1060, #1061, #1086, #1097

Rebuilds:

- Rebuild buildenv modules with correct permissions on wrappers (PR #1101)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Mattias …]

Progress update: software layer (2/2)



Builds for Intel Ice Lake + Intel Cascade Lake CPU targets:

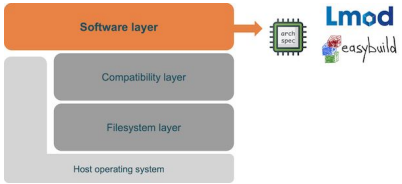
- PR #1071, PR #1072, PR #1071, PR #1078, PR #1079, PR #1080, PR #1082, PR #1084
- **Intel Ice Lake and Intel Cascade Lake are now in sync with the remaining stacks!** 🎉

Recent PRs for general improvements:

- Add CUDA software check to stack comparison CI (open PR# 1087),
- show `/proc/cpuinfo` and run lscpu in CI workflow (PR# 1105)
- fix cudnn hook for non eessi installation (PR# 1099)
- make sure that Lmod cache file is present for all supported CPU targets (PR# 1111)
- run the `check_missing_installations.sh` script with the EB version from the easystack filename (PR# 1094)
- Added hook to restore filtered deps for LLVM builds (PR# 1093)
- Fix EB4 related warning for hooks (PR# 1089)
- Only check for changes to `eb_hooks.py` in a PR context (PR# 1088)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Helena …]

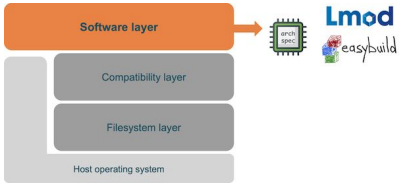
Software layer repository split



- The software-layer repository was split on 11 June 2025
- Main changes
  - Code & scripts that are used to build the EESSI software layer have been relocated to a separate repository: **EESSI/software-layer-scripts**.
  - The minimal `bot/build.sh` script in `EESSI/software-layer` pulls in the latest `main` branch of the `EESSI/software-layer-scripts` repository, symlinks the files in there, and then calls out to the `bot/build.sh` script located in that separate repository.
  - The default branch of the software-layer repository has been changed to `main` (was `2023.06-software.eessi.io`), and houses easystack files for all versions of EESSI (not just `2023.06`).
  - See more details about this change in the support portal: https://gitlab.com/eessi/support/-/issues/139

[Kenneth, Alan …]

Progress update: software-layer-scripts (1/2)

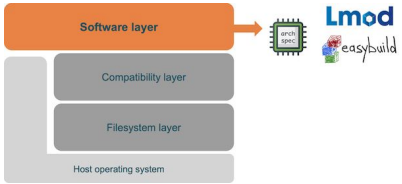


PRs to enable the split (software-layer)

- remove scripts & co that have been relocated to EESSI/software-layer-scripts repository PR #1121 & PR #1122
- update README to reflect changes made when splitting off EESSI/software-layer-scripts repository (PR #1124)
- empty `2023.06-software.eessi.io` branch, add minimal README (PR #1125)
- add minimal `bot/build.sh` script that leverages scripts from EESSI/software-layer-scripts (PR #1126)
- Make sure CI triggers for all workflows on push to main (PR #1129)

[Kenneth, Alan, …]

Progress update: software-layer-scripts (2/2)



PRs to enable the split (software-layer-scripts)

- Remove CI workflow for comparing installed modules across CPU target (belongs in EESSI/software-layer repository) (PR #4)
- Make sure that EESSI version for which build is triggered matches with location of easystack file + fix URL in `create_directory_tarballs.sh` (PR #8)
- Update README + remove CI workflow for verying that README mentions correct EESSI version (PR #12)
- Make sure CI triggers for all workflows on push to `main` (PR #20)

General improvements (**highlights**)

- Use `unionfs` overlay tool (+ debian12 build container that includes it) by default (PR #26) & add support to `eessi_container.sh` for using `unionfs` overlay tool
- Add CI for `EESSI-extend` (PR #15)

Other PRs: #1, #3, #5, #7, #8, #10, #11, #14, #17, #20, #21

[Kenneth, Alan, …]

EESSI pilot repository (RIP)

- `/cvmfs/pilot.eessi-hpc.org`
- The EESSI pilot repository has been unsupported for a long time
  - Loading it would print a deprecation message since about a year
- Recently we added a removal announcement to the init scripts
- **All CernVM-FS servers for EESSI pilot repo have been shut down on 1 July 2025**
  - **The pilot repository is now no longer accessible/available**
- All references to the EESSI pilot repository have been removed from the docs
- See https://gitlab.com/eessi/support/-/issues/141

[Bob]

Integration with Spack

- We've started looking into "integration" of Spack & EESSI
- *How can Spack be used to build & install software on top of what EESSI provides?*
- Initial discussion with Spack developers via `#spack` in EESSI Slack
- EESSI could ship a Spack database that provides details on software included in EESSI
- Proof-of-concept Python script to create Spack DB provided by Todd Gamblin
- Need to work out a concrete use case first, then look into auto-generating Spack DB
- See also https://gitlab.com/eessi/support/-/issues/170

[Kenneth, Alan]

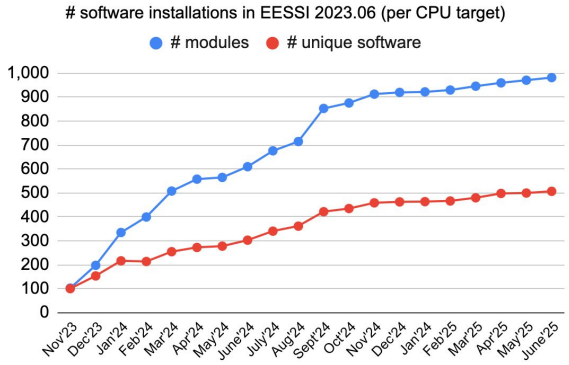
EESSI production repository

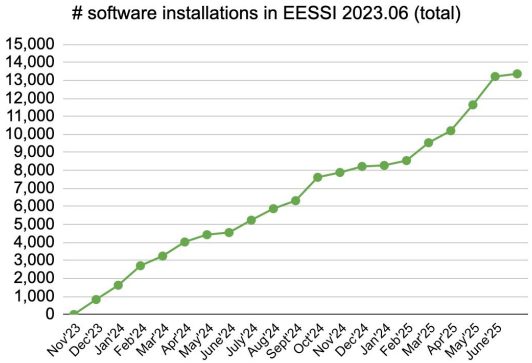
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version 2023.06 is being populated with software via PRs to software-layer repo + build-and-deploy bot

- 13 fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- **Intel Cascade Lake + Ice Lake complete since mid May 2025**
- **Work-in-progress:**
  - **Arm A64FX (for EuroHPC system Deucalion): ~54% done**
- Initial support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
- **Currently: 982 software installations per CPU target**
  (+22 since April'25)
  - **507 different software projects** (+9 since April'25)
    13,361 software installations (across 13+1 CPU targets, +1,721 since Apr'25)
- Current focus:
  - Adding more GPU (CUDA) software installations
  - Processing incoming contributions
  - Complete set of installations for A64FX





EESSI production repository

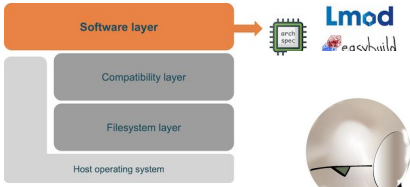
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version 2025.06 will be the next EESSI version

- Compatibility layer is in place
- Init scripts and modules have been prepared:
  https://github.com/EESSI/software-layer-scripts/pull/6
- Waiting for EasyBuild 5.1.1 to be released
  - Includes some fixes that will make it easier to add new CPU targets later on
- Build container based on Debian 12 and `unionfs-fuse` will be used for building software
- The same set of CPU targets will be supported as for 2023.06
- This version will support newer toolchains (foss 2024a and newer)

Bot for building + deploying software (1/3)

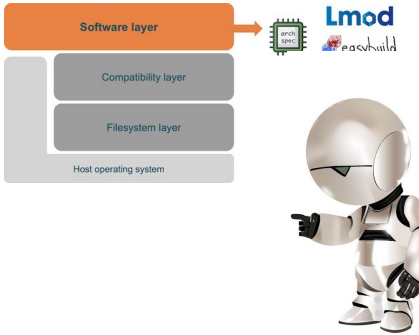


https://github.com/EESSI/eessi-bot-software-layer

- May-June'25
  - PR #300 enable using `ssh` (keys) for obtaining PR contents (supporting private repositories)
  - PR #311 determine test coverage
  - PR #315 make bot chattiness configurable
  - PR #316 use Ubuntu 24.04 in CI
  - PR #318 & PR #319 release v0.8.0
  - PR #321 & PR #322 add CI to build and publish smee-client Docker container image
  - PR #324 make space before bot command optional (`bot: CMD` -> `bot:CMD` or `bot: CMD`)
  - PR #325 revise README.md and app.cfg.example
- Ongoing work
  - Open PRs (change filter logic, adapt arch_target_map & support different tasks)
  - Improvements to ingestion procedure ... see next slide(s)

[Bob, Caspar, Kenneth, Lara, Pedro, Sam, Thomas]

Bot for building + deploying software (2/3)

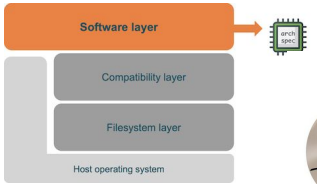


https://github.com/EESSI/filesystem-layer

- (April-)May-June'25
  - Work towards bundling tarballs into a single staging PR
  - Background
    - One tarball per CPU(+GPU) build job, CPUs: 6 -> 8 -> 10 -> 14 * #GPUs ...
    - One staging PR per tarball -> lots of manual clicking ... difficult to spot errors ... not much automatic checking (or difficult to achieve)
  - Current procedure
    - Scan S3 bucket for tarballs, download tarball, metadata file & signatures
    - In staging repo create branch+PR for tarball, used for housekeeping of ingestion state
    - Clever method, so merging PR denotes tarball as approved, closed PR -> rejected

[Bob, Thomas]

Bot for building + deploying software (3/3)



https://github.com/EESSI/filesystem-layer

- Main design ideas for bundling tarballs (or tasks) in a single* PR
  - scan S3 bucket for metadata files instead of tarballs (later, scan for tasks: add, delete, ...)
  - (*) accommodate for batches/sequences each bundling some tarballs, but not all
  - combine target repository, PR number and sequence as branch name
  - restructure staging repo directory layout
    - pointer from metadata file to directory storing information about the staging PR
    - staging PR directory: `target_repository/PR_number/SEQ_number/metadatafile/...`
- Challenges
  - concurrent processing of a tarball by ingestion script & maintainer merging/closing PR
  - correct order of creating initial PR and then adding new tarballs to it
- Draft PRs: PR 213 & PR 232 (213 is initial draft, 232 is improved version)

[Bob, Thomas]

EESSI documentation

eessi.io/docs - GitHub repo github.com/EESSI/docs

Improvements to the EESSI documentation: ~36 merged PRs (+ 6 updates to software overview)

Highlights:

- EESSI webinar training sessions

  (updates to docs related to the sessions, slides and links to presentations were added)
- Add new CPU targets nvidia/grace, Cascade Lake and Ice Lake to software overview
- Blog posts
  - Progress on AMD ROCm: https://www.eessi.io/docs/blog/2025/06/30/rocm/ and https://www.eessi.io/docs/blog/2025/05/26/rocm/
- Documentation on test-suite (added new test, added API docs, https://dashboard.eessi.io)
- Plus other minor changes to the documentation

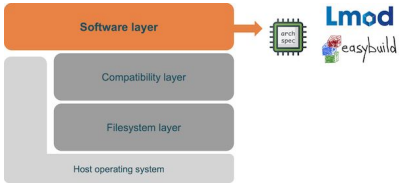
[Lara, Pedro, Helena, Bob, Xin, Kenneth, Caspar…]

software.eessi.io & co as URLs

- Name of CernVM-FS repositories *look* like URLs, but aren't actually
- https://software.eessi.io now redirects to EESSI documentation
  - Same for `http://`
  - Goes to https://www.eessi.io/docs/repositories/software.eessi.io
- Done for all 3 EESSI CernVM-FS repositories
  - https://software.eessi.io
  - https://dev.eessi.io
  - https://riscv.eessi.io

[Alan, Kenneth]

EESSI test suite



eessi.io/docs/test-suite - github.com/EESSI/test-suite

ReFrame

Highlights

- Added test for OpenFOAM (#243)
- Automatically generated API docs (docs: #319)
  - Note: API docs are generated based on docstrings.

    Those should still be cleaned up, but it's a start!
- Various smaller fixes/changes
- New release 0.7.0 and 0.7.1, see github.com/EESSI/test-suite/releases

[Caspar, Sam, Satish, Kenneth, Lara]

EESSI test suite - dashboard

Performance view: https://dashboard.eessi.io

Save to JPG | Save to PNG

10414 records retrieved based on test selection:
7669 records with valid "perf" Y value
2745 records of failed tests
retrieval took 11.561s
280 records left after filtering

Test selection

Start date 03/04/2025 Clear
End date dd/mm/yyyy (optional)

Week | Month | 3 Months | 6 Months | Year

Study	*	(1)
Group	EESSI (164793)	(1)
Test	GROMACS (27706)	(13)

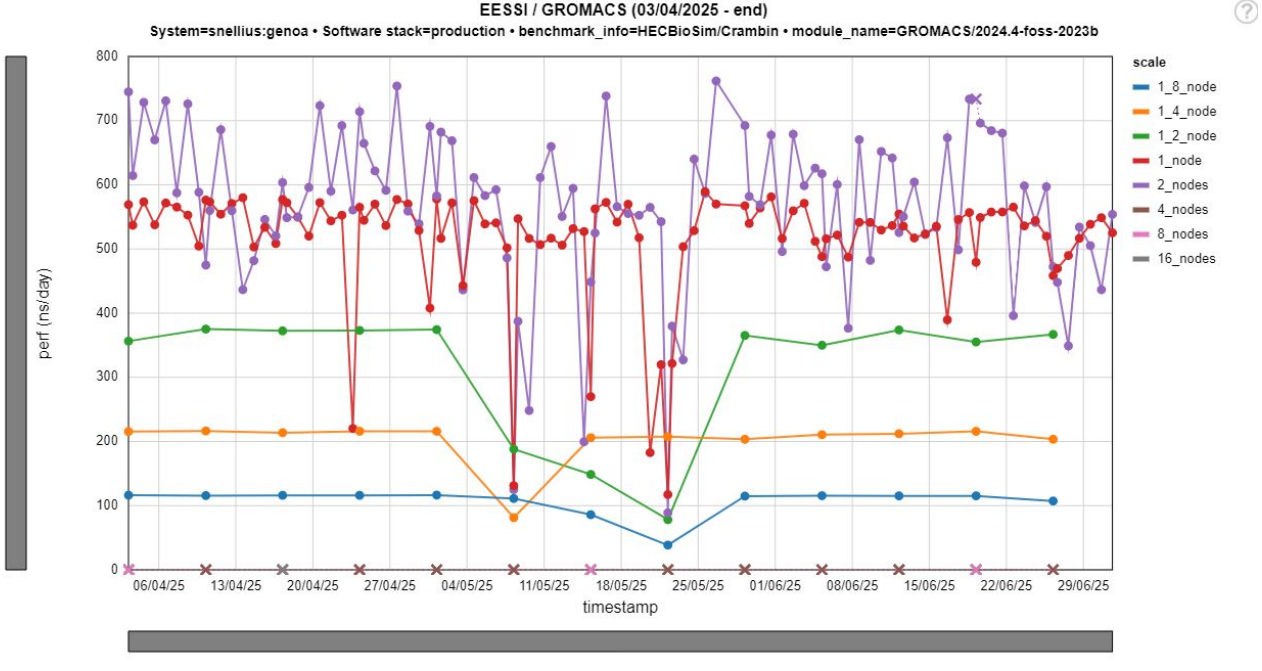
Plotting

X	Time	(7)
Y	perf	(1)
Color	scale	(8)
Shape	(unspecified)	(8)

Nice | Zero | Log

Filtering

System	snellius:genoa	(22)
Environment	default	(1)
Software Stack	production	(1)
benchmark_info	HECBioSim/Crambin	(1)
module_name	GROMACS/2024.4-foss-2023b	(4)
nb_impl	*	(2)
scale	*	(10)



[Caspar, Sam, Satish, Kenneth, Lara]

EESSI test suite - dashboard

Tile-based view: https://dashboard.eessi.io/module-matrix

Click a row to see the module versions

	BETZY: normal	**FRAM:** bigmem	**FRAM:** normal	**Magic Castle:** aarch64 generic 16c 32gb	**Magic Castle:** aarch64 neoverse N1 16c 32gb	**Magic Castle:** aarch64 neoverse V1 16c 32gb	**Magic Castle:** x86_64 generic 16c 30gb	**Magic Castle:** x86_64 haswell 16c 30gb	**Magic Castle:** x86_64 skylake 16c 30gb	**Magic Castle:** x86_64 zen2 16c 30gb	**Magic Castle:** x86_64 zen3 16c 30gb	**SAGA:** bigmem	**SAGA:** normal	**accelgor:** accelgor	**doduo:** doduo	**donphan:** donphan	**gallade:** gallade	**hortense:** cpu_milan	**hortense:** cpu_milan_rhel9	**hortense:** cpu_rome	**hortense:** cpu_rome_512	**joltik:** joltik	**karolina:** qcpu	**shinx:** shinx	**skitty:** skitty	**snellius:** genoa	**snellius:** gpu	**snellius:** gpu_A100	**snellius:** gpu_H100	**snellius:** rome	**vega:** cpu	**vega:** gpu
CP2K				2/2	2/2	2/2	0/2	0/2	2/2	2/2	2/2		0/1		1/2	0/2	0/2	1/2	0/2	1/2	1/2		3/5	0/2	0/1	8/8				8/8	0/5	
ESPResSo				3/6	3/6	3/6	3/6	3/6	3/6	3/6	3/6		0/3		3/6	3/8	3/6	3/6	3/3	3/6	3/6		3/6	2/6	0/3	23/24				23/24	0/6	
GROMACS				13/13	13/13	9/13	13/13	13/13	13/13	13/13	13/13		0/3	0/2	4/6	2/8	4/6	3/6	3/6	3/6	2/6		9/22	1/6	0/3	17/30				24/30	6/22	2/2
LAMMPS				4/4	4/4	4/4	4/4	4/4	4/4	4/4	4/4		0/2	2/2	4/4	4/8	2/4	4/4	4/4	4/4	4/4		6/10	2/4	0/2	16/16				16/16	0/10	2/2
MetalWalls				2/2	2/2	2/2	2/2	2/2	2/2	2/2	2/2		0/1		2/2	2/2	2/2	2/2	2/2	2/2	2/2		0/5	1/2	0/1	4/8				5/8	0/5	
OSU-Micro-Benchmarks	2/2	2/2	4/4	5/6	6/6	5/6	5/6	5/6	6/6	6/6	5/6	2/2	4/8	1/3	8/8	8/18	8/8	8/8	8/8	8/8	8/8	0/3	6/22	4/8	0/4	22/28	6/6	0/8	0/8	22/28	0/22	8/8
QuantumESPRESSO				7/8	7/8	7/8	4/8	4/8	8/8	8/8	7/8		0/2		2/4	1/4	1/4	0/4	3/4	1/4	0/4		16/20	1/6	2/4	14/32				31/32	10/20	
TensorFlow				3/3	3/3	3/3	0/3	0/3	3/3	3/3	3/3		0/1		1/2	2/2	1/2	1/2	1/2	1/2	1/2		3/9	0/2	0/1	9/12				9/12	0/9	

[Caspar, Sam, Satish, Kenneth, Lara]

Improving NVIDIA GPU support

- Supported CPU/GPU architecture combinations

CPU microarchitecture			CUDA CC 7.0	CUDA CC 8.0	CUDA CC 9.0
x86_64	generic				
	intel	haswell			
		skylake			
		cascadelake	N		
		icelake		N	
		sapphirerapids			
	amd	zen2		N	
		zen3		N	
		zen4			N
aarch64	generic				
	neoverse_n1				
	neoverse_v1				
	nvidia	grace			N

- N = native build, the rest is cross-compiled
- See also https://github.com/EESSI/docs/pull/504

[Caspar]

Improving NVIDIA GPU support

- EasyBuild has CUDA sanity check since v5.1.0.
  Checks if installed binaries:
  - Have device code + PTX code for the configured CUDA Compute Capability
  - Don't have device / PTX code for *unrequested* architectures

  (see also https://blog.easybuild.io/2025/05/27/easybuild-5.1.0/#cuda-sanity-check)

- Default EasyBuild configuration: no failures, just warnings.

- Planned default EESSI: failure for missing device/PTX code, warning for additional device/PTX code.
  - Missing device code (and no compatible PTX): fix build behaviour
  - Missing PTX or unrequested device code: fix build behaviour *or* demote to warning (i.e. not a blocker!)

- Future plan: exploit NVCC_APPEND_FLAGS or NVCC_PREPEND_FLAGS

[Caspar]

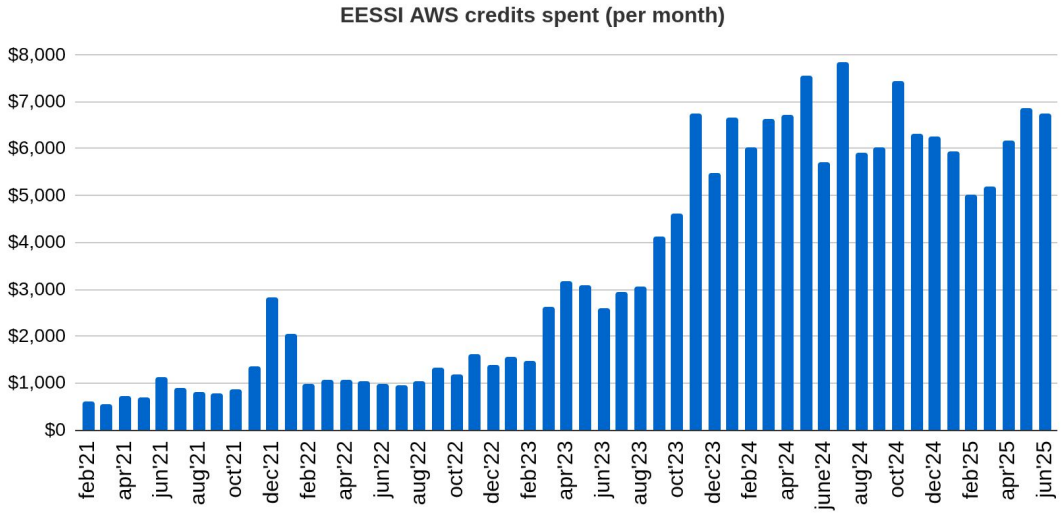
Adding support for AMD GPUs (ROCm)

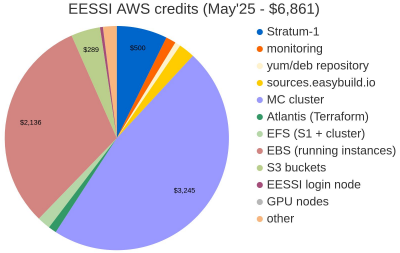
- Work on adding support for AMD GPUs is well underway
- Built on the back of much better LLVM support in EasyBuild
- Status updates covered in recent EESSI blog posts:
  - Mapping the AMD ROCm Ecosystem
  - Building ROCm Support in EESSI
- Funded effort by Microsoft Azure via INUITS to help out with this effort
  - Weekly meetings held (Alan/Kenneth)
- **Can now build ROCm 6.4.x on top of EESSI + run the ROCm validation suite**
- Deployment will be done via a dev.eessi.io repository initially to figure out structure, …

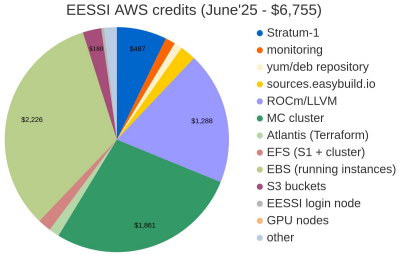
[Bob, Alan, Kenneth, …]

Sponsored AWS credits

- Ask in #aws-resources Slack channel to get access!
- ~$6,861 + $6,755 “spent” in May'25 + Jun'25 on Stratum-1 servers, monitoring, demos, sources.easybuild.io,Slurm clusters (build bot), **dev.eessi.io builds**
- **Bulk of consumed credits due to EESSI build-and-deploy bot**
- Regular sync meetings with AWS - more sponsored credits have been provided for coming months



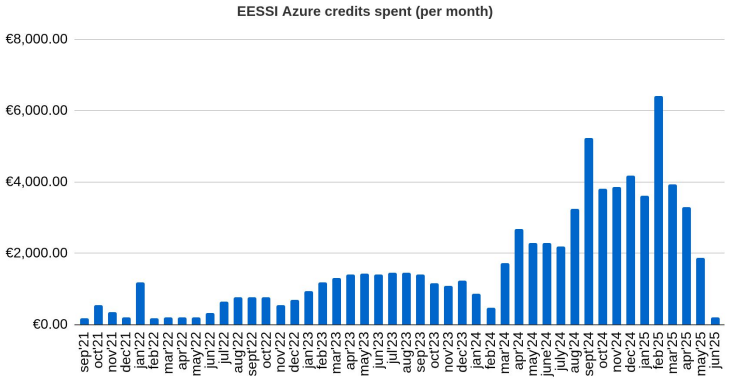


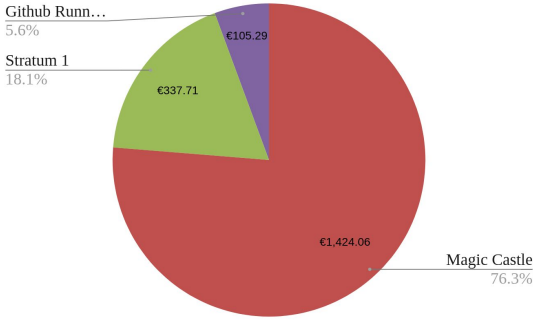


[Kenneth, Bob]

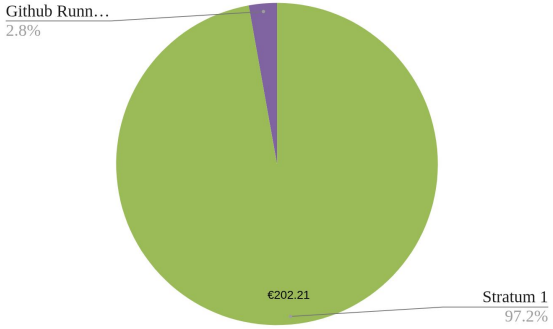
Sponsored Azure credits

- Ask in #azure-resources Slack channel to get access!
- In May+June'25: €1,868+€209 worth of credits spent
- Used for: Stratum-1 mirror server
  - **Slurm build cluster for dev.eessi.io has been moved to AWS**
  - Costs of keeping Stratum-1 up are OK until end of 2025 (via SURF)
- Discussions to extend sponsored credits have stalled…





May'25 (€1,868)



Jun'25 (€209)

[Kenneth, Bob]

EESSI Governance - (interim) Steering Committee

- (interim) Steering Committee https://eessi.io/docs/governance
  - Kenneth/Lara, Alan/Davide, Bob/Henk-Jan, Caspar/Satish, Thomas/Terje
- Most recent meeting was held 19 June 2025
- Actively discussing governance for EESSI
  - Formalize/document how we collaborate in EESSI community
  - See WIP PR: https://github.com/EESSI/docs/pull/456
- Initial meeting with Linux Foundation Europe on 10 June 2025
  - Typically only cover software, not services

EESSI webinar series

- 5 webinars in a row, Mondays in May/June 2025
- All webinars recorded and available at https://eessi.io/docs/training/2025/webinar-series-2025Q2
- Total of 168 registrations
  - 75% academic registrations
  - Active participants from ~35 (EESSI as a base) to ~60 (Intro to EESSI)
- Considering making the series an annual thing

EESSI CI/CD hackathon after EuroHPC User Days

- Hackathon session in wake of EuroHPC User Days 2025
  https://eurohpc-ju.europa.eu/media-events/events/eurohpc-user-days-2025-2025-09-30_en
  (Tue 30 Sept + Wed 1 Oct in Copenhagen)
- Hands-on, helping developers/admins get introduced to EESSI, onboard their software, …
  - Likely to have focus on EuroHPC Centres-of-Excellence
  - Exact program TBD
- Being organised in collaboration with EuroHPC (CASTIEL2 and NCC Denmark)
- More information soon via https://eessi.io/docs/training/2025/hackathon-eurohpc-user-days

[Kenneth, Alan]

EESSI Happy Hour sessions

A main theme across several sessions. Each session will include:

- Quick updates: Recent EESSI developments + highlights of upcoming events

- Spotlight topic (~15 min)

- A focused introduction to the day's topic
- May feature:
  - A live demo
  - A guest speaker sharing insights

- Open discussion

- Time for questions, ideas, and community sharing

Planned to start mid August 2025 on Mondays 14:00-15:00 CEST

[Richard]

Next EESSI update meetings

- EESSI update meetings are **bi-monthly**
- First Thursday of the month at 14:00 CE(S)T, only in odd months
  - iCalendar URL for calendar integration
- Next meetings:
  - Thu 4 Sept 2025 14:00 CEST (12:00 UTC)
  - Thu 6 Nov 2025 14:00 CET (13:00 UTC)
  - Thu 8 Jan 2026 14:00 CET (13:00 UTC)